YIZHOU DING

Singapore | +65 89824608 | DING0205@e.ntu.edu.sg | yizhouding1.github.io

EDUCATION

Nanyang Technological University (NTU) Singapore
MSc in Financial Engineering *2026 – Present*
- **Coursework (completed / in progress):** Stochastic Calculus in Finance, Derivative Securities, Financial Time Series Analysis, Exotic Options & Structured Products, AI Techniques in Finance, Numerical Methods for Financial Instrument Pricing.

Southwestern University of Finance and Economics (SWUFE) & University of Delaware (UD) Chengdu, China
Dual Bachelor's Degrees in Financial Mathematics & Finance; GPA: 90.75/100 *2022 – 2026*
- **Coursework:** Linear Algebra, Probability Theory, Mathematical Statistics, ODE, Stochastic Processes, Econometrics.

RESEARCH INTERESTS

AI for Economics and Finance; Applied Economic Theory; Asset Pricing; Decentralized Finance; Numerical Methods.

RESEARCH EXPERIENCE

Global Institute of Finance, Technology, and Society (GIFTS), NTU Singapore
Research Assistant to Prof. Lin William Cong *Aug. 2026 – Present*

AI-Agent-Based Financial Market Simulation with Structural Equilibrium
- Solved the benchmark all simulations target: the exact Bayesian–Nash equilibrium of a multi-group Kyle (1989) call auction, nesting Hellwig (1980), by homotopy and damped Newton for 19 investor compositions $\times$ 2 conducts, verified by five independent checks.
- Replaced the solver with agents that never see the equilibrium: (i) recursive-least-squares learners reach it in all 38 cells ($<0.1\%$ of gains forgone); (ii) 20 LLM traders, told only the rules, profit-linked pay, and their own record, settle within 3% of its depth.
- Computed four policy equilibria ex ante (forced seller, stamp duty, short-sale fee, liquidity withdrawal); the LLM market reaches each (3 other LLMs agree), pinning effects that vary 4–30$\times$ in published ABMs (tax elasticity -1.38, in the empirical range).

Financial AI Benchmarks: Economic Decisions and On-Chain Execution
- Design a benchmark testing whether LLM agents choose economically sound on-chain actions, not merely execute given ones, across 4 DeFi decision domains $\times$ 4 difficulty levels, scored by counterfactual mainnet-fork execution rather than LLM judges.
- Built the lending-domain foundation for Aave V3, Compound III, and Morpho Blue from contract code.

AI for Prediction-Market Design and Dispute Prevention (with Prof. Mingzhe Li, Great Bay University)
- Pilot study: rebuilt Polymarket's on-chain history (2.38M markets, oracle disputes, clarifications) from a Polygon archive node.
- From launch-time text alone (Mantel–Haenszel odds ratios, 407K markets), contingency clauses cut dispute odds by $\sim$70% and "announced/confirmed" triggers raise upheld disputes 2.3$\times$; built a matched, time-split benchmark for LLM pre-launch screening.

Nanyang Business School, NTU Singapore
Research Assistant to Prof. Samia Badidi; project: how AI reshapes sell-side analysts' careers *Sep. 2026 – Present*
- Mapped 63 target analyst titles onto Revelio's 17K-category role taxonomy (777 kept), each cross-checked against O*NET.
- Built the WRDS employment panel (SQL/Python): 397M raw records filtered by role, industry (NAICS-52, RICS), seniority, intern, and outlier rules into 12.7M positions of 8.2M people at 316K finance firms (2010–2026).
- Audited role labels against raw titles via live WRDS queries: 16% mismatch for the 17K-category label vs. 81% for the 1.5K one.

School of Mathematics, Southwestern University of Finance and Economics Chengdu, China
Research Assistant to Prof. Pin Lyu *Sep. 2024 – Present*
- **Manuscript:** Ding, Y., Zhou, X., & Lyu, P. "A Variable-Step Crank–Nicolson and High-Order Compact Scheme for a Nonlinear Black–Scholes Type Equation." *Under review.*
- Designed, with coauthors, a linearized scheme for derivative pricing with counterparty risk and funding costs (variable-step Crank–Nicolson in time, fourth-order compact in space) that needs one linear solve per step, without nonlinear iteration.
- Proved unconditional stability and $\mathcal{O}(\tau^2 + h^4)$ convergence (energy method), including under negative rates; verified rates in MATLAB; led manuscript preparation.

INDUSTRY EXPERIENCE

Shenyin Wanguo (SWS) Futures, Research Institute, Quantitative Research Intern, Shanghai *Jul. 2025 – Sep. 2025*
- Tested whether macro data explain China's 10-year government bond yield beyond Nelson–Siegel factors: the residual is persistent and Granger-caused by CPI and macro surprises; rolling XGBoost on 30 macro variables explained 52% of it out of sample.
- Conducted an event study of four extreme ETF-option rallies (2015–2024) on lead–lag, volatility, and tail losses; drafted reports for the institute's quant series on yield-curve models and algorithmic stablecoins.

Galaxy Futures, Derivatives Business Headquarters, Derivatives Research Intern, Shanghai *Jan. 2025 – May 2025*
- Tested whether intraday gamma gains of long straddles offset theta decay on minute-level data for 10 Chinese commodity options: buying at each product's best 5-min entry and selling 30 min later earned $\sim$80% on crude oil at a 10% allocation (2021–23) and 741.5% annualized for the 10-option portfolio (2022–23, before costs); published as a two-part article on the firm's WeChat.
- Mapped intraday volatility seasonality in $\sim$20 commodity futures; the straddle paid off for night-session crude oil, not PTA.
- Built a local LLM (DeepSeek-R1-7B) pipeline forecasting rebar futures direction from forum text, price–volume data, and an analyst framework; selective prediction (abstaining on low-confidence calls) raised the hit rate from 57.5% to 60.0% over 279 trading days.
- Compared hedges for rebar procurement: an option collar cut cost variance by 97.7%, against 41.7% for a futures hedge.

TEACHING

Teaching Assistant, MISY330: Database Design & Implementation (Prof. Leting Zhang, UD) *Feb. 2026 – Jun. 2026*

SKILLS & HONORS

Programming: Python, MATLAB, R, C++, SQL
Languages: Mandarin (native); English (IELTS 7.5; GRE 326)
Honors: Academic Scholarship (Top 10%); Leadership Scholarship (2022-2026);

REFERENCE

Prof. Fei Xie, Chaplin Tyler Professor of Finance; Associate Chair, Department of Finance, University of Delaware | xief@udel.edu